

Foundation (Jalapeno)

- Q1.** The graph of $y = x^2 + 4x + 4$ has a vertex at:
 (A) (0, 4)
 (B) (2, 0)
 (C) (-2, 0)
 (D) (0, 0)
 (E) (1, 2)
- Q2.** A basketball player's jump height is modeled by $y = -2x^2 + 12x + 1$. What is the axis of symmetry?
 (A) $x = 2$
 (B) $x = 3$
 (C) $x = 4$
 (D) $x = 5$
 (E) $x = 6$
- Q3.** For the parabola $y = x^2 - 6x + 8$, what are the x -intercepts?
 (A) (2, 0), (4, 0)
 (B) (0, 0), (6, 0)
 (C) (1, 0), (8, 0)
 (D) (2, 0), (6, 0)
 (E) (4, 0), (8, 0)
- Q4.** A soccer player kicks a ball with an initial velocity of 25 m/s. The height of the ball is modeled by $y = -5x^2 + 25x$. What is the maximum height?
 (A) 50 m
 (B) 62.5 m
 (C) 75 m
 (D) 80 m
- (E) 100 m
- Q5.** The graph of $y = 2x^2 - 4x - 1$ opens:
 (A) Up
 (B) Down
 (C) Left
 (D) Right
 (E) None of the above
- Q6.** A cyclist travels a distance of $y = 2x^2 + 5x + 1$ meters in x seconds. What is the direction of the parabola?
 (A) Up
 (B) Down
 (C) Left
 (D) Right
 (E) None of the above
- Q7.** For the parabola $y = x^2 - 2x - 6$, what is the y -intercept?
 (A) (0, 6)
 (B) (0, -6)
 (C) (0, 2)
 (D) (0, -2)
 (E) (0, 1)
- Q8.** The axis of symmetry for $y = -x^2 + 2x + 3$ is:
 (A) $x = -1$
 (B) $x = 0$
 (C) $x = 1$
 (D) $x = 2$
 (E) $x = 3$

Open Ended Questions (FRQ)

- Q9.** Graph the parabola $y = x^2 + 2x - 3$ and find the vertex, axis of symmetry, and x -intercepts.

- Q10.** A skateboarder's jump height is modeled by $y = -x^2 + 8x + 2$. Find the maximum height and the time it occurs.

Chef's Tips

- Provide additional support for students struggling with graphing parabolas.
- Encourage students to use a variety of methods to solve problems, such as factoring and the quadratic formula.
- Remind students to label their graphs clearly and include all necessary information, such as axis labels and titles.